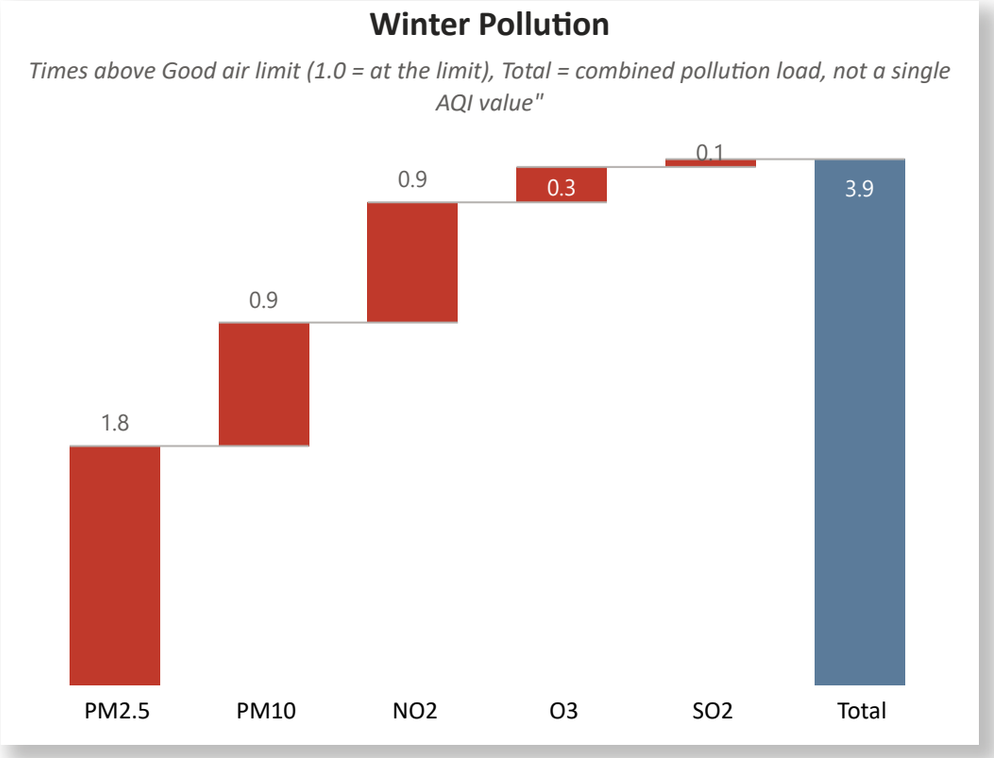
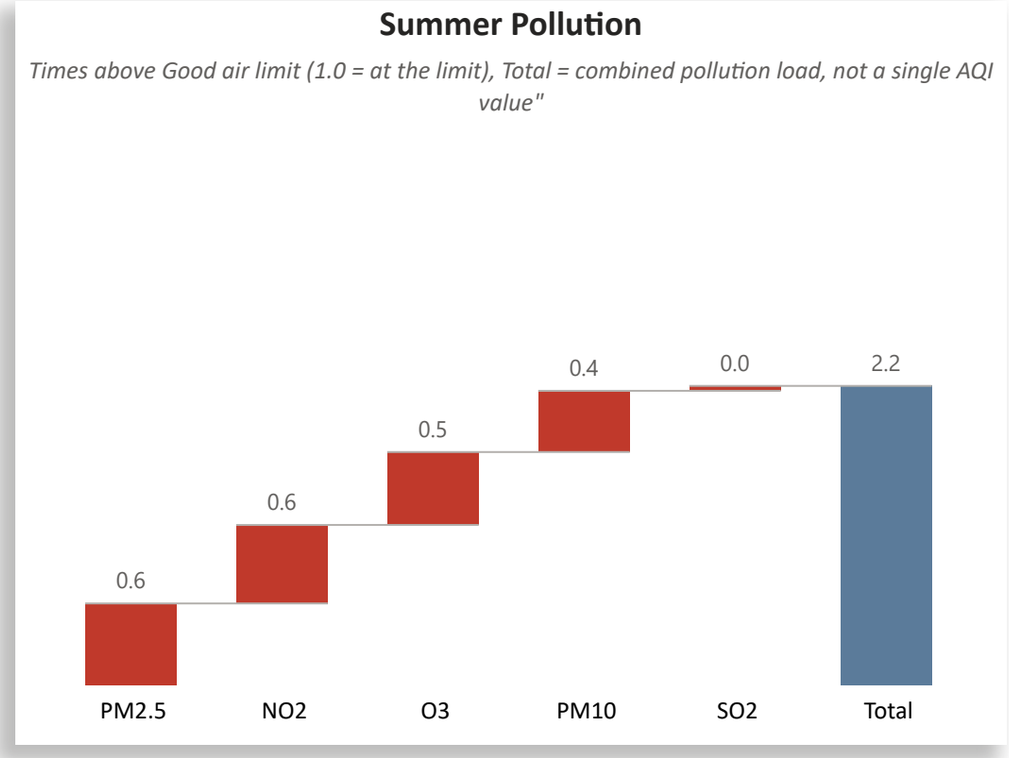
Ingredients of Bad Air by Season



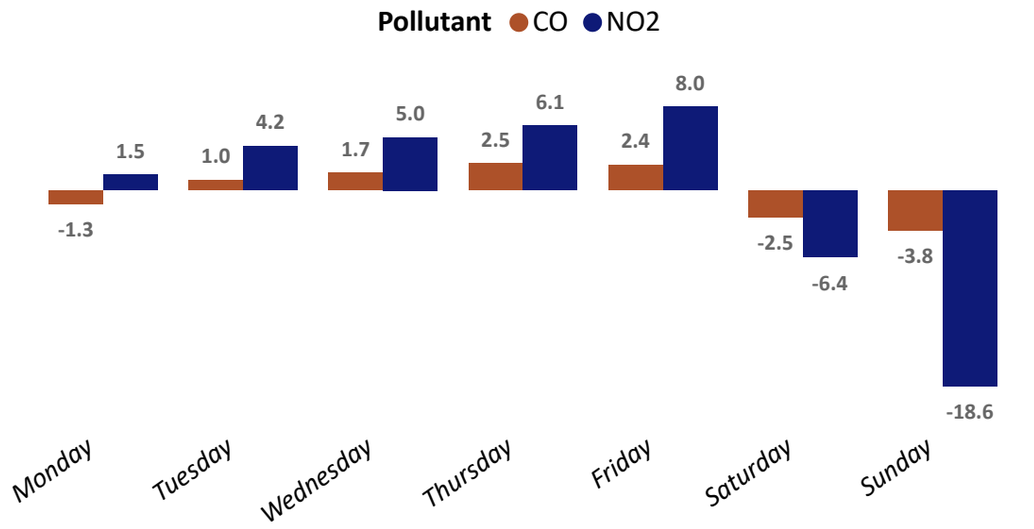


Pick a City

- ○ Dubrovnik
- ○ Osijek
- ○ Rijeka
- ○ Split
- ○ Vis
- ◉ Zagreb

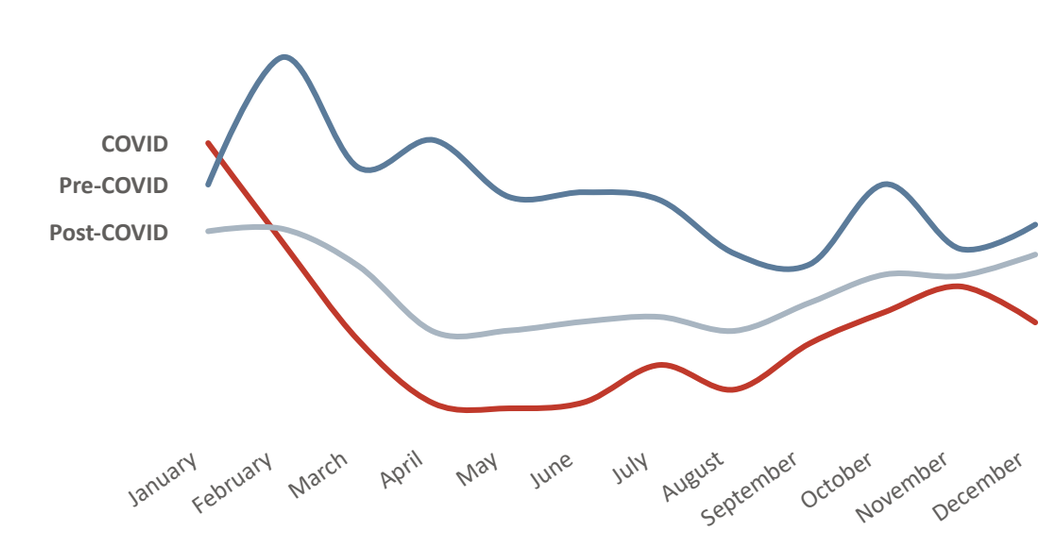
Weekdays Pollute More / Weekend Brings Relief

Each day's average shown as % above or below the overall mean

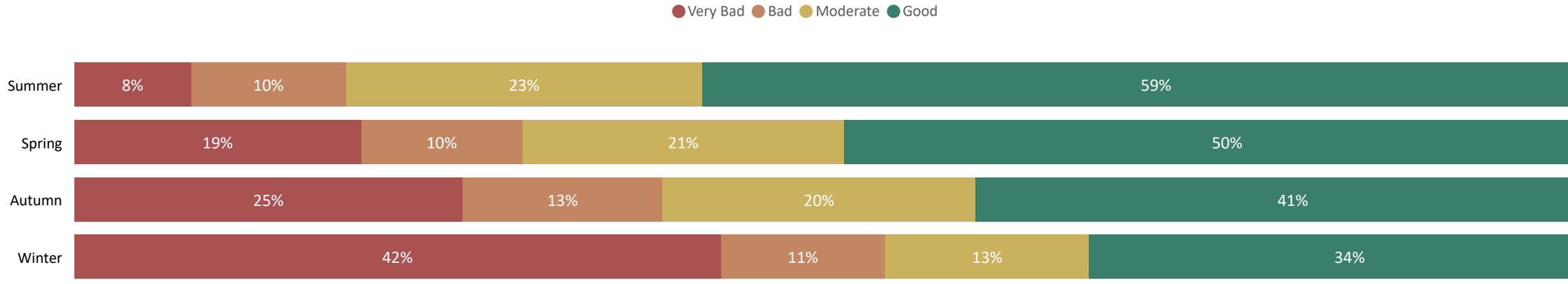


COVID Scared the Pollution Away - And it never Came Back

Monthly NO_2 levels (µg/m³) - Lockdown (2020) vs Pre-COVID (2019) vs. Post-COVID (2021-2024)



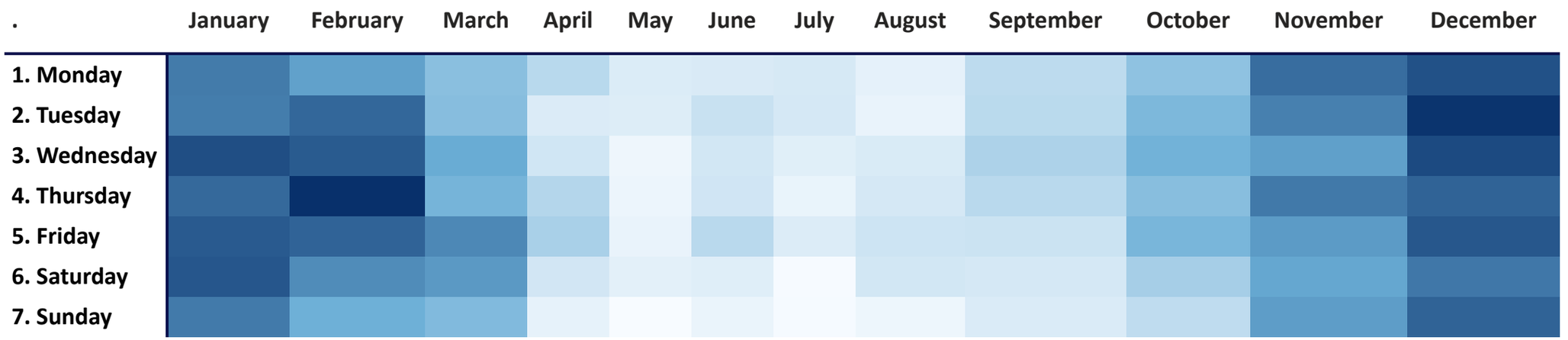
Percentage of Hours Spent Breathing Unhealthy Air, by Season



Dubrovnik | Osijek | Rijeka | Split | Vis | Zagreb

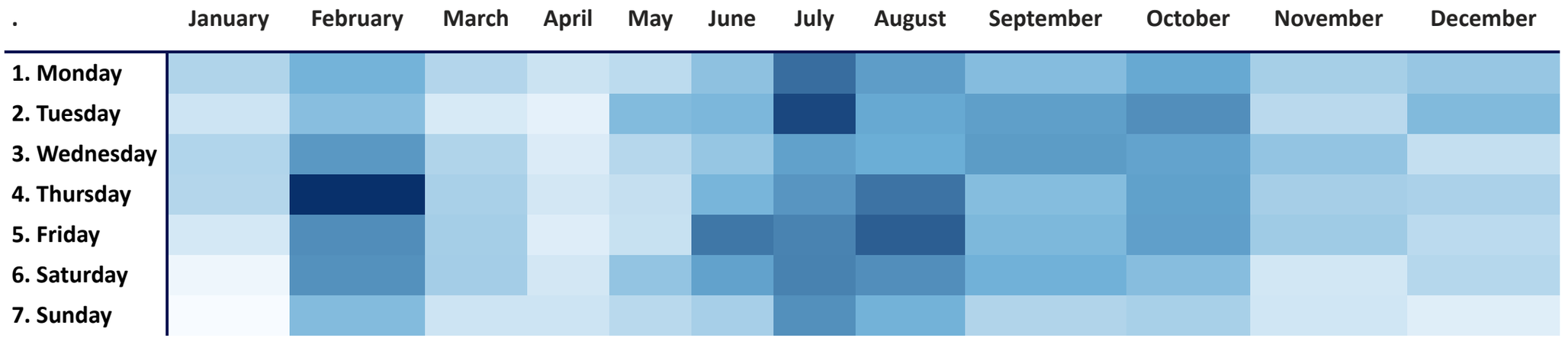
Zagreb's Winter Problem

Darker = poorer air quality (PM10 PM2.5 NO2 SO2)



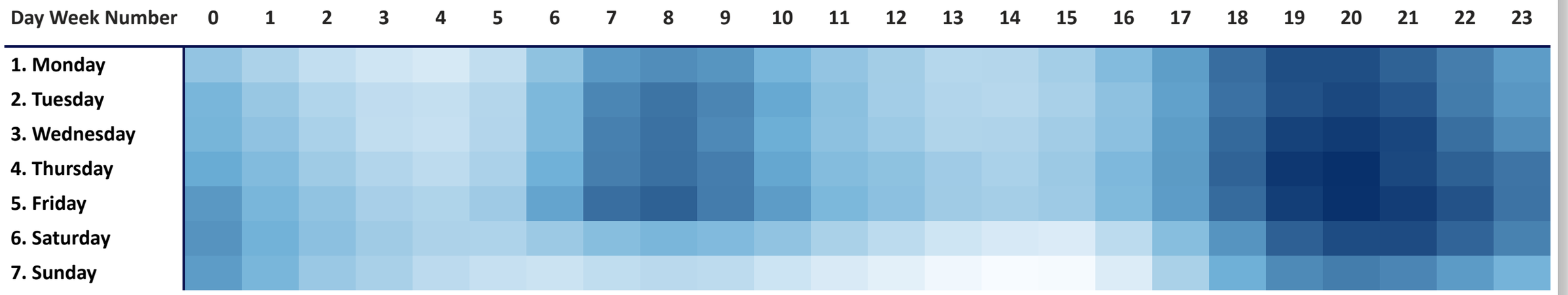
Split Pollutes in Reverse

Darker = poorer air quality (PM10 PM2.5 NO2 SO2)



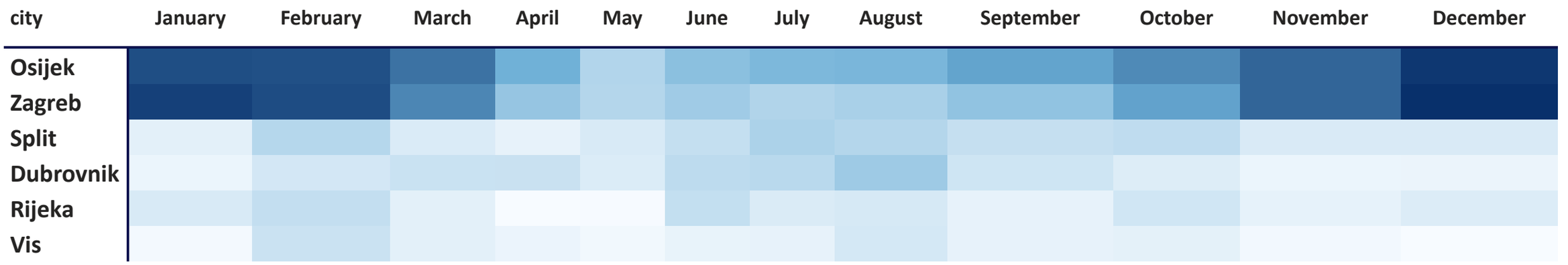
Mornings and Evenings Hit Hardest

Darker = poorer air quality (PM10 PM2.5 NO2 SO2)

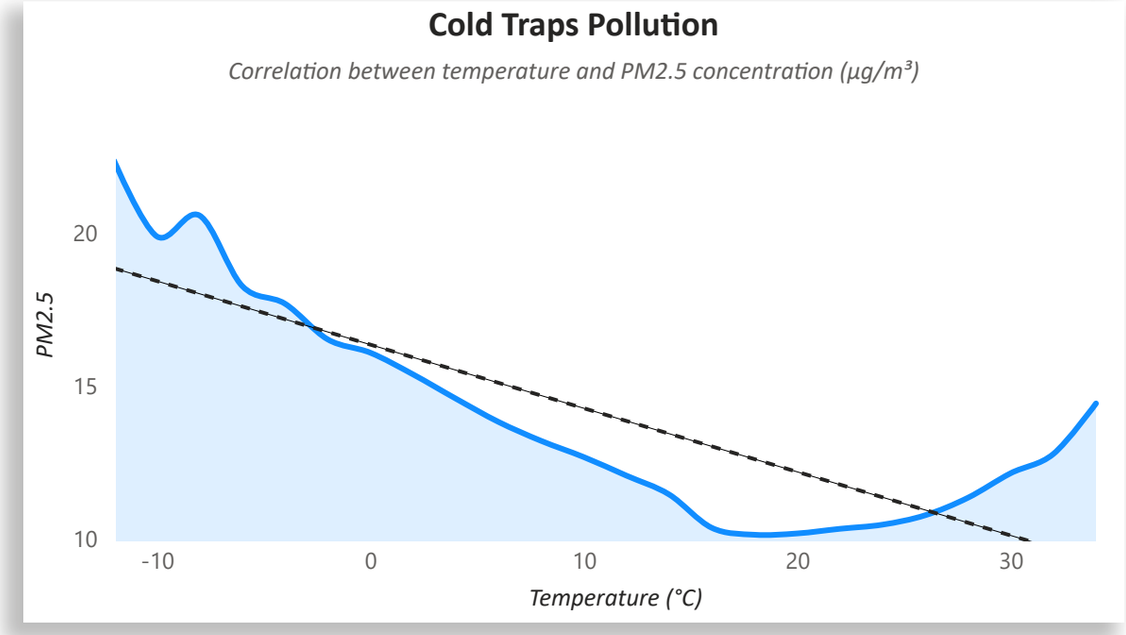


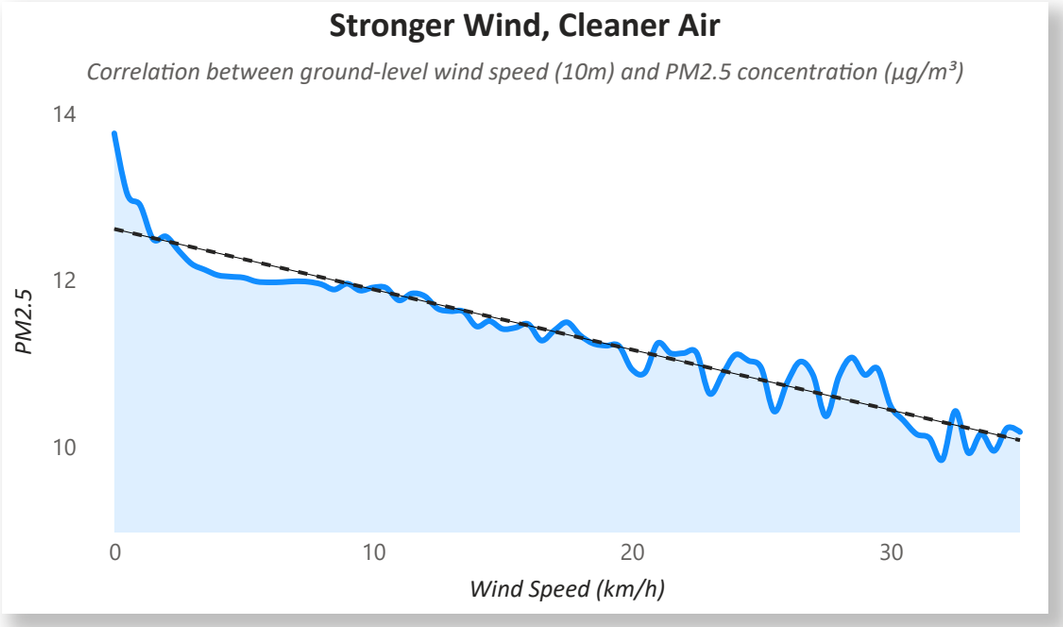
Where Heating Season Hits Hardest

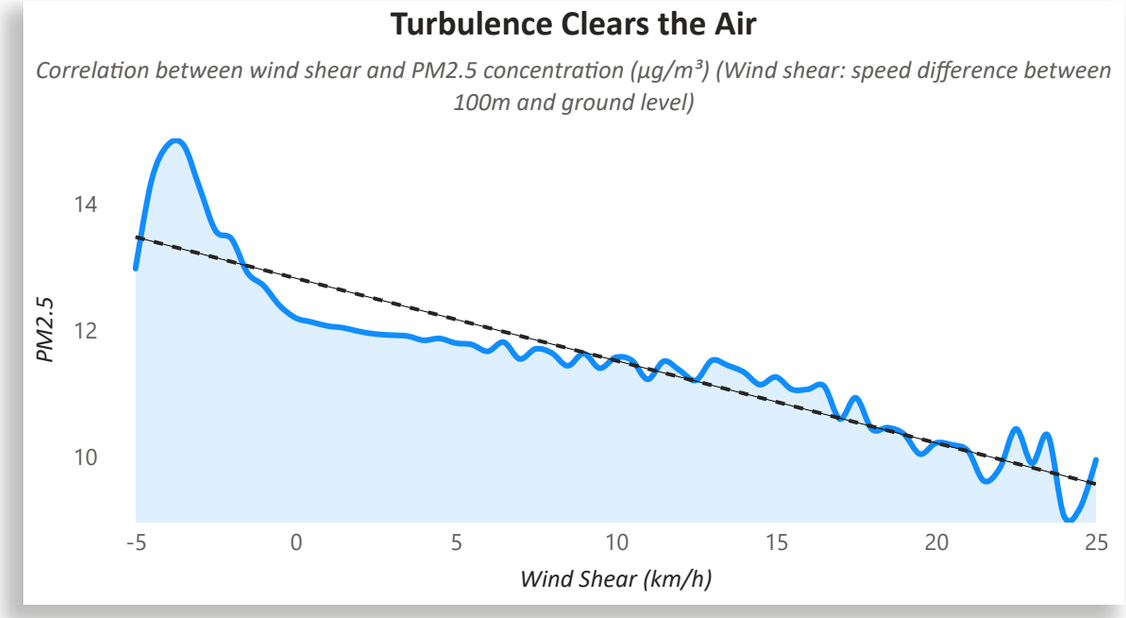
Darker = poorer air quality (PM10 PM2.5 NO2 SO2)



What Scares PM2.5 Away?

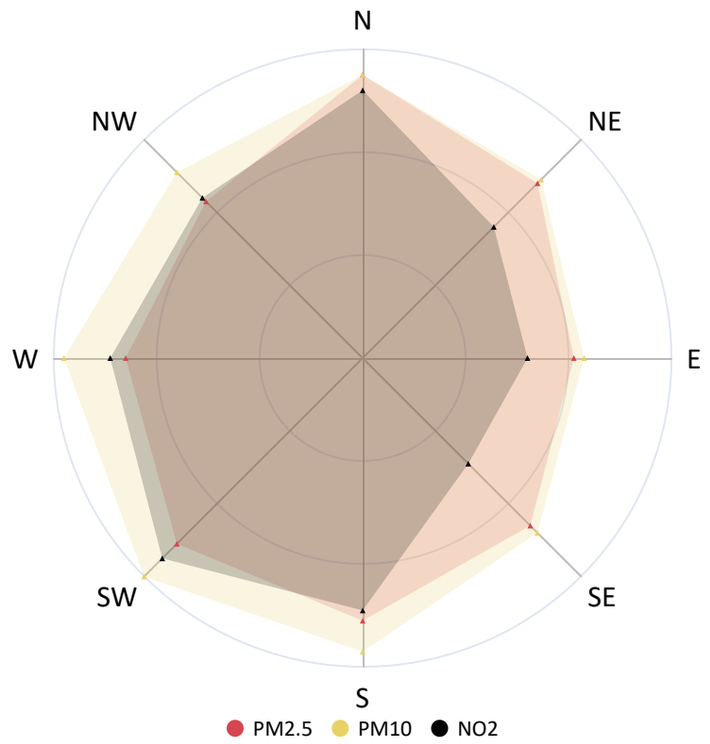






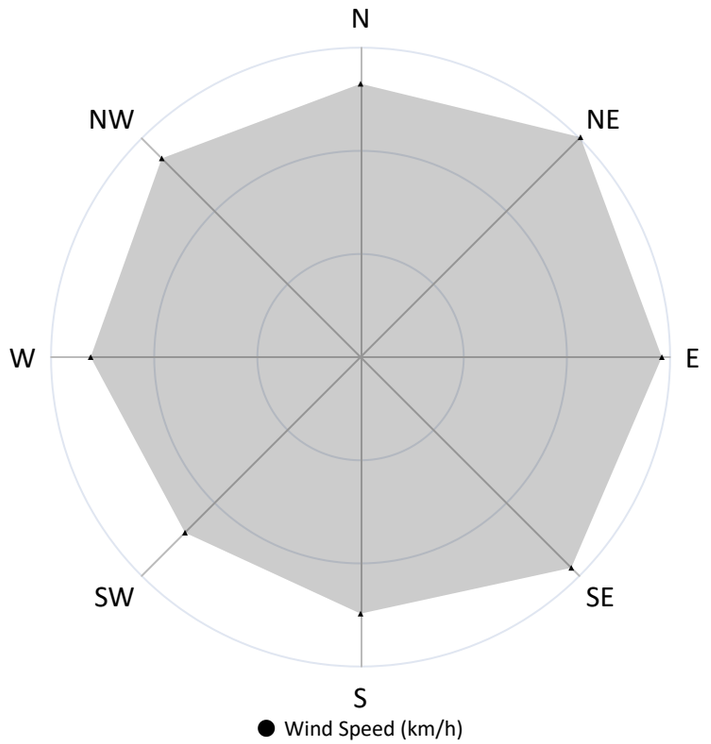
Which Wind Direction Brings the Dirtiest Air

Pollution ratio by wind direction - 1.0 = EAQI Good limit, SMALLER AREA = CLEANER AIR



Wind Power by Direction

km/h - stronger wind typically disperses pollution



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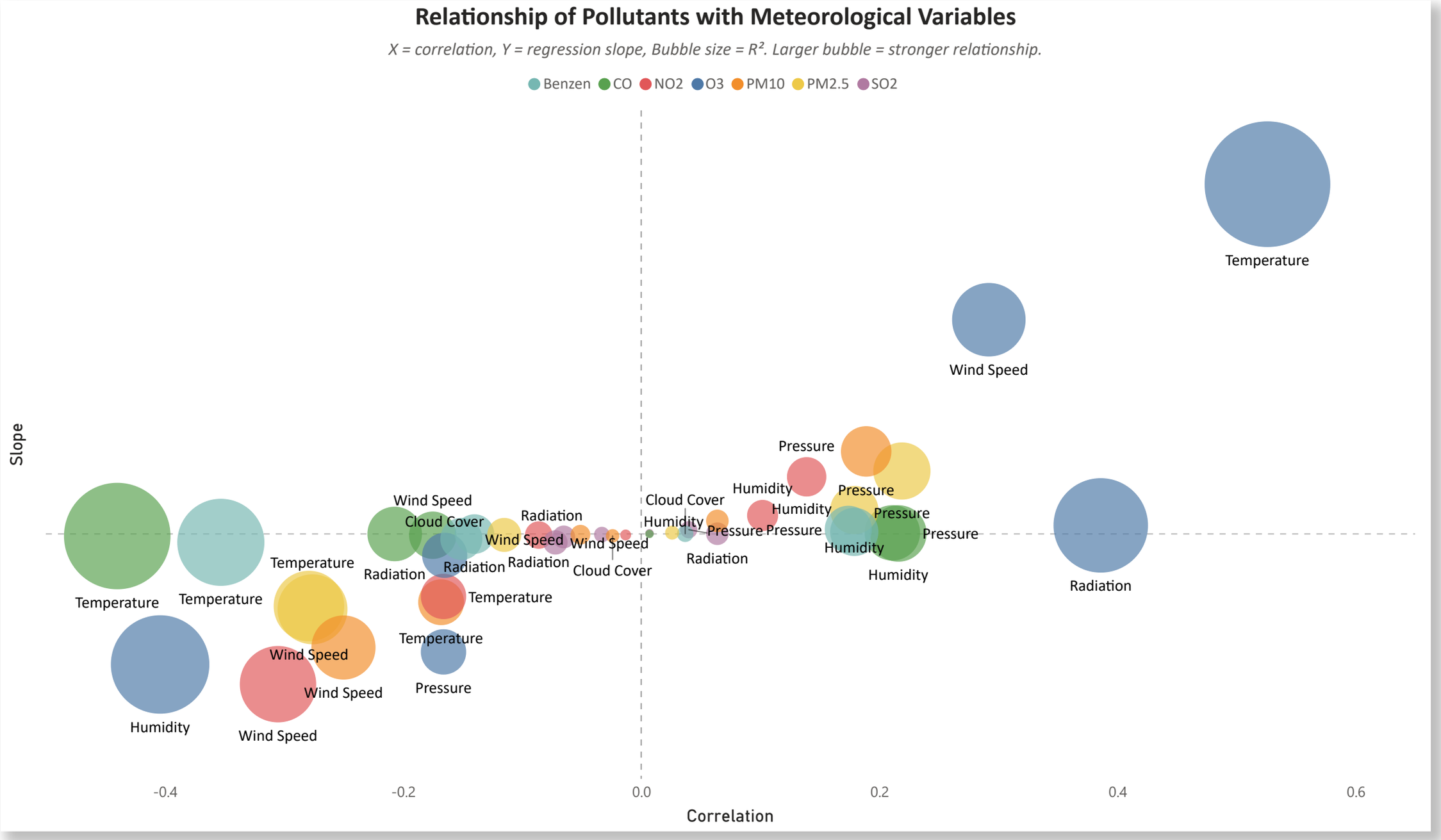
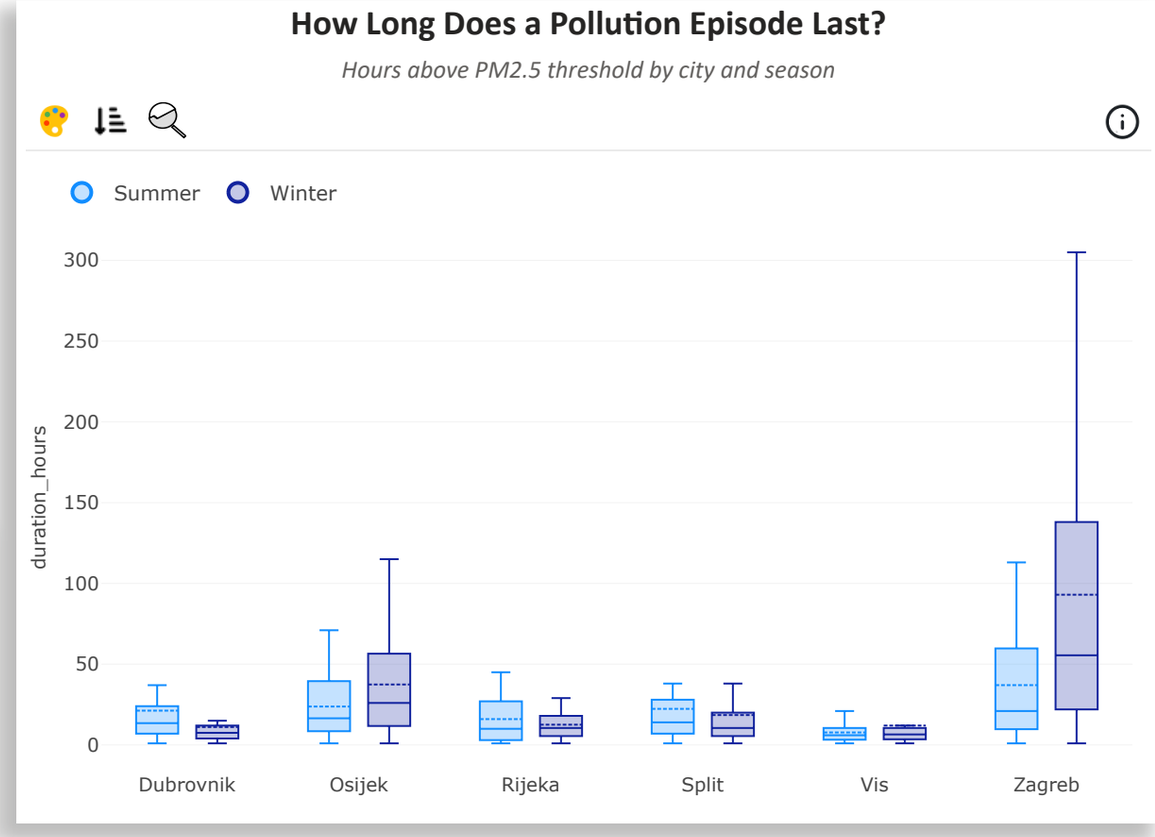
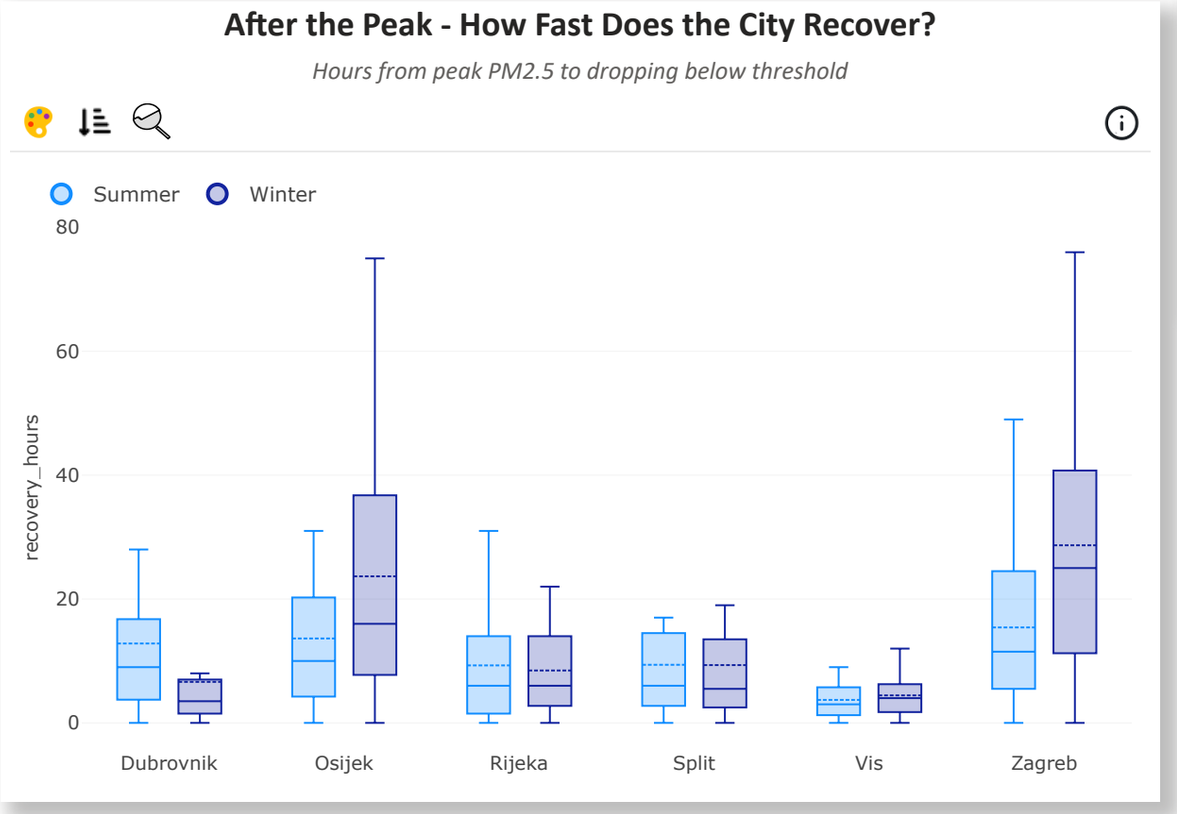
- **Split**
- Vis
- Zagreb


Relationship of Pollutants with Meteorological Variables
X = correlation, Y = regression slope, Bubble size = R². Larger bubble = stronger relationship.
Benzen
CO
NO2
O3
PM10
PM2.5
SO2
Slope
Correlation
-0.4
-0.2
0.0
0.2
0.4
0.6
Temperature
Wind Speed
Pressure
Humidity
Radiation
Cloud Cover

How Long Pollution Episodes Last and How Fast Cities Recover





Pick the line -What counts as bad air?

Pollution Threshold

PM2.5 (µg/m³)

- ◉ 20
- ○ 25
- ○ 50